

NBA Player Performance Analysis

Problem Statement

The goal of this project is to analyze the factors influencing NBA player performance during the 2023-2024 regular season and playoffs. The analysis aims to identify key performance metrics that significantly impact player effectiveness. The insights derived will provide actionable recommendations to inform strategic decisions, player training, and game strategies.

Context

In the highly competitive NBA, understanding what drives player performance is crucial for teams aiming to optimize strategies, enhance training programs, and improve overall team effectiveness. This project will provide data-driven insights that can help teams and analysts better understand player effectiveness in both the regular season and the playoffs.

Criteria for Success

The project will be considered successful if it uncovers key performance metrics that offer actionable and valuable insights for NBA teams, analysts, and coaches. These insights should help optimize player performance and improve strategic planning during games.

Scope of Solution Space

The analysis focuses on player data from the 2023-2024 NBA season, covering both the regular season and playoffs. This project will use statistical analysis and data visualization techniques to uncover patterns, trends, and performance drivers. The goal is to isolate individual player contributions, despite team dynamics, using the available data.

Constraints

Key constraints include potential issues with data quality, challenges in isolating individual contributions from team dynamics, and the necessity of preprocessing the data (e.g., handling missing values, normalizing performance metrics).

Stakeholders

The primary stakeholders include NBA teams, coaches, and analysts who will use the findings to make strategic decisions, optimize training programs, and improve team performance.

Data Sources

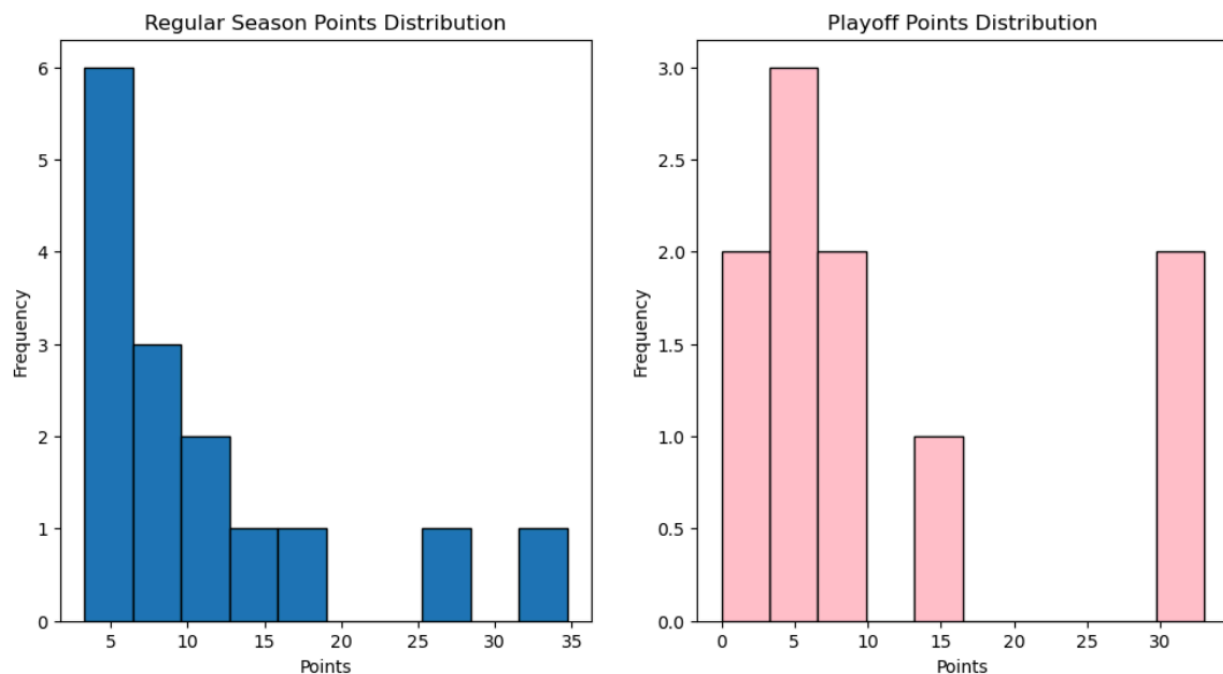
The dataset used for this analysis is NBA Player Stats Season 23-24 from Kaggle. It contains detailed statistics on player performance during both the regular season and playoffs, including metrics such as points, assists, rebounds, and shooting percentages.

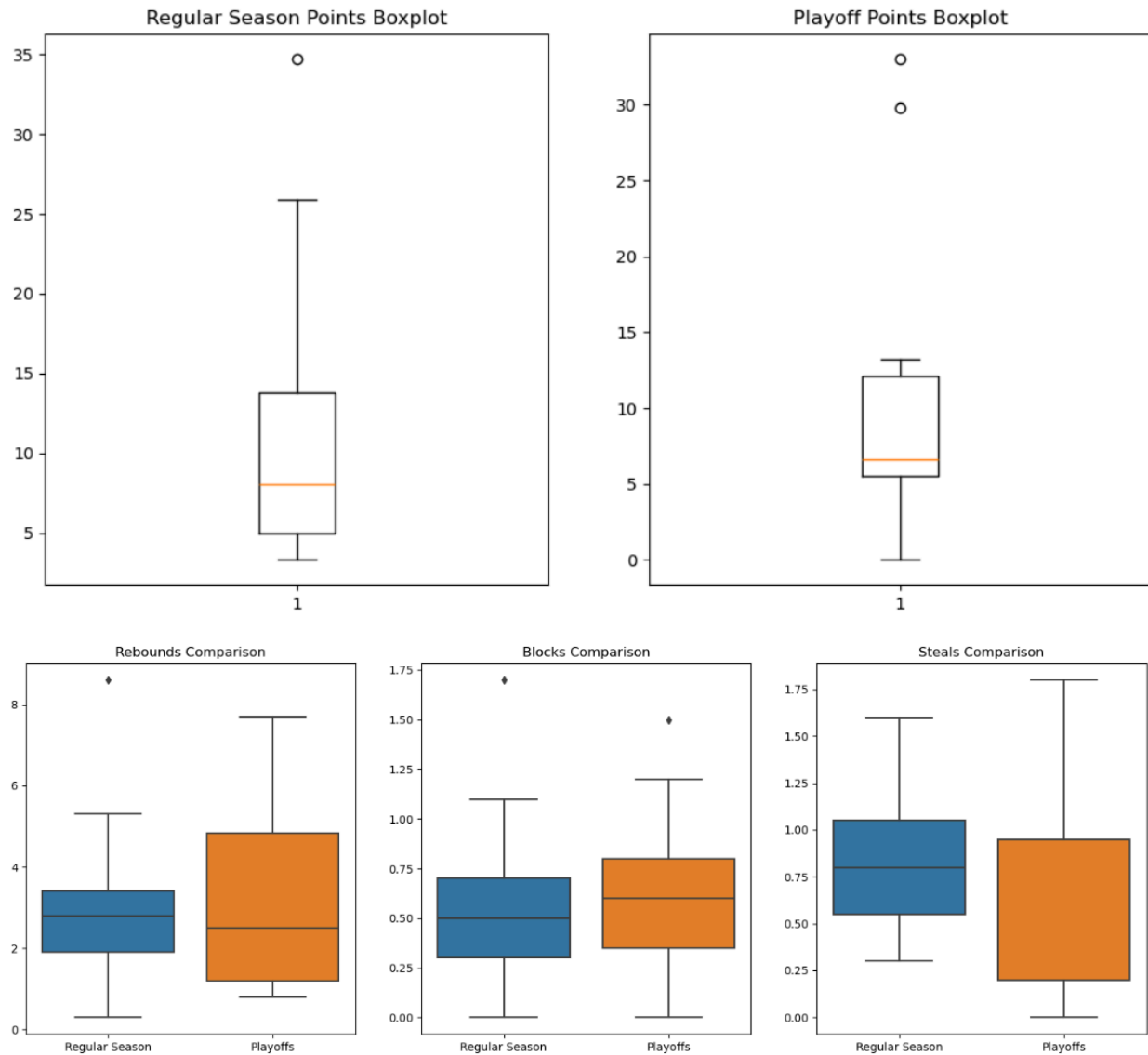
Exploratory Data Analysis (EDA)

For EDA, we focused on the performance of players from the Philadelphia 76ers during the regular season and playoffs. Key findings include:

- **No significant statistical differences** were found in defensive attributes like rebounds, blocks, and steals between the regular season and the playoffs.
- Key players such as Joel Embiid and Tyrese Maxey maintained or increased their contributions, while Tobias Harris showed a decline in points during the playoffs.
- **Overall team performance** in the regular season and playoffs did not show any major differences.

EDA Visualization





Modeling Overview

The primary objective of the modeling phase was to predict NBA player performance based on key metrics from both the regular season and the playoffs in the 2023-2024 NBA season. Three models were employed to approach this problem:

- Linear Regression
- Random Forest Regressor
- Gradient Boosting Regressor

Data Splitting

The data was split into training and test sets for both regular season and playoff games:

- Regular Season: The dataset was split with an 80-20 train-test ratio.
- Playoffs: The playoff dataset followed the same 80-20 split pattern.

1. Linear Regression

Linear Regression was used as a baseline model to understand the relationship between player performance metrics and effectiveness.

Results:

- Regular Season:
 - MSE: 1.4016
 - R-squared: 0.9818
- Playoffs:
 - MSE: 9.4382
 - R-squared: 0.9497

Linear Regression performed exceptionally well on the regular season data, capturing over 98% of the variance. While performance in the playoffs was slightly lower, it still achieved a strong R^2 value of 0.95.

2. Random Forest Regressor

The Random Forest model, an ensemble-based method, was employed to handle complex relationships between variables.

Results:

- Regular Season:
 - MSE: 39.1977
 - R-squared: 0.4913
- Playoffs:
 - MSE: 107.4397
 - R-squared: 0.4276

Despite being a powerful model, Random Forest underperformed compared to Linear Regression, with lower R-squared values and higher MSE for both the regular season and playoffs.

3. Gradient Boosting Regressor

Gradient Boosting was explored as an alternative model, known for handling small datasets and noisy data more effectively.

Results:

- Regular Season:
 - MSE: 28.0834
 - R-squared: 0.6355
- Playoffs:
 - MSE: 14.9114
 - R-squared: 0.9206

Gradient Boosting outperformed Random Forest, especially for the playoff data, achieving an R^2 score of 0.92. However, it still lagged behind Linear Regression in terms of regular season performance.

In the model comparison, Linear Regression outperforms others in the regular season with an MSE of 1.4016 and an R-squared of 0.9818, while also showing strong playoff performance (MSE of 9.4382, R-squared of 0.9497). Random Forest has poor performance (MSE: 39.1977, R-squared: 0.4913 in the regular season; MSE: 107.4397, R-squared: 0.4276 in the playoffs), and Gradient Boosting shows moderate results (MSE: 28.0834, R-squared: 0.6355 in the regular season; MSE: 14.9114, R-squared: 0.9206 in the playoffs).

Linear Regression emerged as the best-performing model for both the regular season and playoffs based on R-squared and MSE values. Gradient Boosting was the second-best model, particularly for the playoff data. Random Forest underperformed, suggesting that simpler models like Linear Regression may be more effective when the dataset has a clear linear relationship.

Conclusion and Recommendations

- Key Performance Drivers: Based on the analysis, metrics like points, assists, and shooting percentages were found to be strong predictors of player performance.
- Modeling Insights: Linear Regression, the simplest model, performed the best, indicating that player performance during the regular season and playoffs can be explained by relatively straightforward relationships between metrics.
- Strategic Recommendations: NBA teams and analysts can use these insights to focus on training and improving key performance areas like shooting accuracy and assist-making, which have a significant impact on player effectiveness.
- Future Research: Further studies could explore additional factors such as player fatigue, game conditions, and psychological aspects that might affect performance.